

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering & Research Ravet, Pune An Autonomous Institute NBA Accredited (4 UG Programs) NAAC A++ Accredited ISO 21001:2018 Certified IQAC PCCOER	
Academic Year: 2025-26 Term-I	Formative Assessment	ACAD/R/11-FA

Department: BCA

Class: FY

Div: A&B

Subject: Mathematical Foundation of Computer Science-I

Maximum Marks: 20

Duration: 1 hr

Subject Code: CC-101-CA

Date:

17/11/2025

19
20

Note: 1. Attempt all Questions

2. Give explanation or justification wherever required.

3. Neat diagrams must be drawn wherever necessary.

Course Outcomes:

CO No.	Course Outcomes	BT Level
CC101-1.1	Develop a basic understanding of fundamental mathematical concepts using set theory.	BT 2
CC101-1.5	Demonstrate problem solving skills using the concept of matrix operations.	BT 3

Question No.	Question	CO/ BTL	Marks
Q.1	Attempt any Two (Unit 1)		10 Marks
A	Prove the De Morgan Law for sets using set-builder notation and logical equivalences $\overline{A \cap B} = \bar{A} \cup \bar{B}.$	CC 101-1.1 / BTL: 3	5
B	Show that the set of all integers is countable.	CC 101-1.1 / BTL: 3	5
C	i) What is the Cartesian product $A \times B \times C$, where $A = \{0,1\}, B = \{1,2\}$, and $C = \{0,1,2\}$ ii) What is the power set of the set $\{0,1,2\}$?	CC 101-1.1 / BTL: 2	5
Q.2	Attempt any Two (Unit 2)		10 Marks
A	Explain the Types of Matrices.	CC 101-1.5 / BTL: 2	5

	B	Find $P + Q, P - Q$ for $P = \begin{bmatrix} 1 & 0 & 4 \\ -1 & 2 & 2 \\ 0 & -2 & -3 \end{bmatrix}$ $Q = \begin{bmatrix} -1 & 3 & 5 \\ 2 & 2 & -3 \\ 2 & -3 & 0 \end{bmatrix}$	CC 101-1.5 / BTL: 3	5
	C	If matrix $T = \begin{bmatrix} 3 & -4 & 5 \\ 2 & 6 & 0 \\ -1 & 2 & -3 \end{bmatrix}$ then find its determinant.	CC 101-1.5 / BTL: 3	5

*****END*****

Department Seal



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Academic Year: 2025-26
Term-I

Formative Assessment

ACAD/R/11-FA

Department: Computer Engg.
Subject: Computer Architecture
Subject Code: CC-102T-CA

Class: FY

Maximum Marks 20

Div:

Duration: 1 hr

Date: 21/11/2025

Note: 1. Attempt all Questions

2. Give an explanation or justification wherever required.

3. Neat diagrams must be drawn wherever necessary.

Course Outcomes:

CO No.	Course Outcomes	BT Level
CC-102.1	Apply Boolean algebra to simplify complex logic circuits and convert between different number systems (binary, decimal, hexadecimal) to solve real-world computing problems.	BT-3
CC-102.2	To construct Logic Gates and their circuits to perform logical addition, subtraction, and multiplication.	BT-4, 6

Two Question		CO / BTL	Marks
Q1	Attempt any Three ^{Two} (Unit 1)		10 Marks
	A Convert BCD to Excess-3 Code & Binary to Gray Code	CC-102.1	5 M
	B Convert 1. $(11A.62)_{16} = (?)_{10}, (?)_8, (?)_2$ & 2. $(3DB)_{16} = (?)_8, (?)_{10},$	CC-102.1	5 M
	C Convert 1. $(792)_{10} = (?)_8, (?)_{16}, (?)_2$ & 2. $(268)_{16} = (?)_8, (?)_{10},$	CC-102.1	5 M
	D Convert 1. $(1001.011)_2 = (?)_{10}, (?)_8, (?)_{16}$ & 2. $(FBCD)_{16} = (?)_{10}, (?)_8$	CC-102.1	5 M
Q2	Attempt any Three ^{Two} (Unit 2)		10 Marks
	A Convert Binary code to Gray code (1) 11011011 (2) 1100 (3) 10101 (4) 01001 (5) 110001	CC-102.2	5 M
	B Explain 1's complements and 2's complements. Also, solve $(11011)_2$ using both methods	CC-102.2	5 M
	D Perform Binary Subtraction 1. $(9)_{10} - (4)_{10} = (?)_2$ & 2. $(13)_{10} - (12)_{10} = (?)_2$	CC-102.2	5 M
	E Perform Binary addition 1. $(0011010)_2 + (0001100)_2$ & 2. $(101011)_2 + (111111)_2$	CC-102.2	5 M

*****END*****

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Academic Year: 2025-26 Term-I	Formative Assessment	ACAD/R/11-FA

Department: BCA
 Subject: Problem Solving Technique
 Subject Code: SEC101-CA

Class: FY
 Maximum Marks: 20

Div: A and B
 Duration: 1 hr
 Date: 18/11/2025

- Note:*
1. Attempt all Questions
 2. Give explanation or justification wherever required.
 3. Neat diagrams must be drawn wherever necessary.

CO No.	Course Outcomes	BT Level
SEC-101.1	Understand basic terminology of computers, problem solving, programming language and their evolution	BT1
SEC-101.2	Solve problem requirements by asking questions to disambiguate the requirement statement.	BT2

Question No.	Question	CO/ BTL	Marks
Q1	Attempt Any two (Unit 1)		10 Marks
	A Explain arithmetic operators..	101.1/ BT1	
	B Write C program to find average of three subjects of students	101.1/ BT1	
	C What is an algorithm? Give 1 example.	101.1/BT2	
Q2	Attempt Any two (Unit 2)		10 Marks
	A Write C Syntax, Statement and Expressions	101.2/ BT1	
	B Explain Relational Operators.	101.2/ BT1	
	C Explain Printf() and Scanf() function	101.2/BT3	

*****END*****

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A. Y.:2025-26 Term-I	Formative Assessment	ACAD/R/II-FA

Department: BCA.

Subject: Environmental Science and Sustainability

Subject Code: VAC_101-CA

Class: FY

Maximum Marks: 20

Div: All

Duration: 1 hr

Date: 9/11/2025

20
20

Note: 1. Attempt all Questions

2. Give explanation or justification wherever required.

3. Neat diagrams must be drawn wherever necessary.

Course Outcomes:

CO No.	Course Outcomes	BT Level
VAC- 101.1	To Understanding: Understand basic environmental concepts ,the irrelevance to business operations, and forthcoming sustainability challenges	BT Level 2 — Understanding
VAC- 101.4	To Explain key environmental laws and regulations in India.	BT Level 3 — Understanding

Q. No.	Question			CO / BTL	Marks
(Unit 1)					10 Marks
Q. 1	A	i.	What is Environment?	VAC-01.1/ BTL- 1	1 Marks
		ii.	Write the Major segments of the environment	VAC-01.1/ BTL- 1	2 Marks
		iii.	List any four Historical environmental movements.	VAC-01.1/ BTL- 1	2 Marks
	B	Attempt any One			
		i.	Describe the major components and segments of the environment	VAC-01.1/ BTL- 2	5 Marks
		ii.	Discuss the man–environment relationship.	VAC-01.1/ BTL- 2	5 Marks

(Unit 4)					10 Marks
Q2	A	i.	What is pollution?	VAC-101.4/ BTL- 1	1 Marks
		ii.	List the different types of pollutions (any four)	VAC-101.4/ BTL- 2	2 Marks
		iii.	Give the four major causes of environmental pollution.	VAC-101.4/ BTL- 1	2 Marks
	B	Attempt any One			
		i.	List the different types of pollutions. Explain any one type pollution in details.	VAC-101.4/ BTL- 1&2	5 Marks
		ii.	Describe solid waste management and its importance.	VAC-101.4/ BTL- 3	5 Marks

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Academic Year: 2025-26 Term-I	Formative Assessment	ACAD/R/II-FA

Department: BCA
 Subject: Introduction to Data Science
 Subject Code: CC-101-CA

Class: FY
 Maximum Marks: 20

Div: A&B
 Duration: 1 hr
 Date:
 20/11/2025

14
20

- Note:*
1. Attempt all Questions
 2. Give explanation or justification wherever required.
 3. Neat diagrams must be drawn wherever necessary.

CO No.	Course Outcomes	BT Level
OE 101-1.1	Define and explain the key concepts of data science, including its tasks lifecycle stage.	BT-2
OE 101-1.2	Apply appropriate data pre-processing techniques and use data visualization to explore, interpret and communicate data insight effectively.	BT-2

Question No.	Question	CO/ BTL	Marks
Q.1	Attempt any Two (Unit 1)		10 Marks
	A Describe the Data Science lifecycle with a suitable diagram.	OE 101-1.1/BT-2	5
	B Differentiate between structured, semi-structured, and unstructured data with examples.	OE 101-1.1/BT-2	5
	C List and explain any ^{five} four applications of Data Science.	OE 101-1.1/BT-2	5
Q.2	Attempt any Two (Unit 2)		10 Marks
	A What is the role of statistics in Data Science? Explain briefly.	OE 101-1.2/BT-2	5
	B Define data objects and attributes with suitable examples.	OE 101-1.2/BT-2	5
	C Explain nominal, binary, and ordinal attributes with examples.	OE 101-1.2/BT-2	5

*****END*****

Department Seal